



$$x^2 - 2ax + a^2 - 1 = 0$$

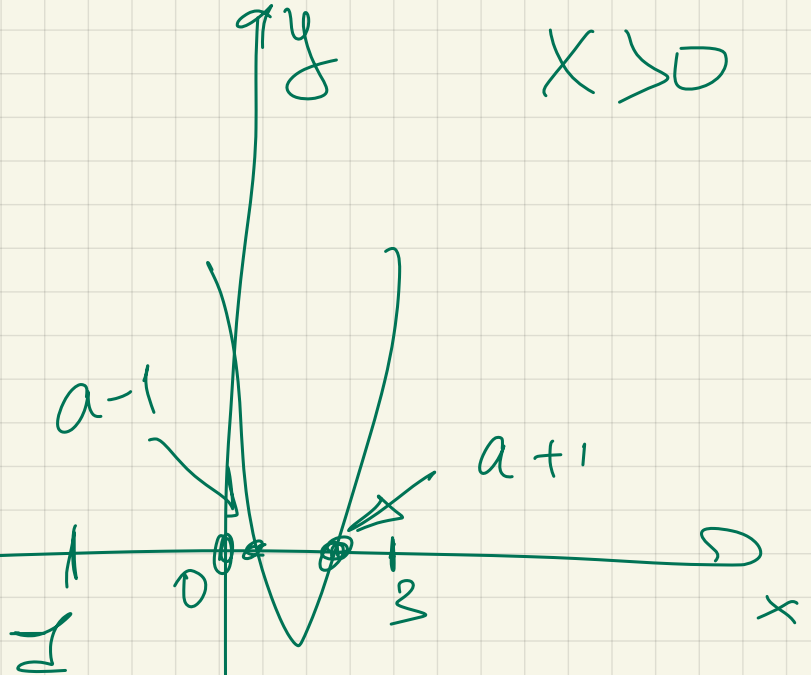
$$D = 0$$

$$(x-a)^2 - 1 = 0$$

$$x > 0$$

$$x = \pm 1 + a$$

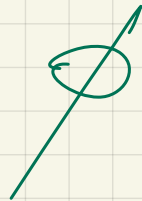
$$\begin{aligned} a+1 &< 3 \\ a-1 &> 0 \end{aligned}$$



$$x^2 - 2ax + a^2 + 1 = 0$$

$$(x-a)^2 + 1 = 0$$

$$(x-a)^2 \neq -1$$



$$\begin{aligned} a &< 2 \\ a &> 1 \end{aligned}$$

$$y = \frac{\sqrt{a+1} - 2\cos 3x + 1}{\sin^2 3x + a + 2\sqrt{a+1} + 2}$$

$$\cos 3x = t \quad -1 \leq t \leq 1$$

$$\sqrt{a+1} = b \geq 0$$

$$\sqrt{a+1} = 0 \quad \underline{a = -1}$$

$$h(t) = \frac{b - 2t + 1}{-t^2 + b^2 - 1 + 2b + 2}$$

$$1 - \cos^2 3x = \sin^2 3x$$

$$a + 1 = b^2$$

$$a = b^2 - 1$$

$$= \frac{b + 1 - 2t}{-t^2 + b^2 + 2b + 2} = \frac{(b + 1) - 2t}{-t^2 + (b + 1)^2}$$

$$\leq b + 1 = c$$

$$h(t) = \frac{c - 2t}{c^2 - t^2 + 1}$$

$$-1 \leq t \leq 1$$

$$0 \leq b = c - 1$$

$$c \geq 1 \quad b = 0$$

$$h(1) = 2$$

$$h(-1) = 3$$

$$-1 \leq t \leq 1$$

$$c \geq 1$$

$$\frac{c-2t}{c^2-t^2+1} = 2$$

$$\frac{c-2t}{c^2-t^2+1} = 3$$

$$c-2t = 2c^2 - 2t^2 + 2$$

$$2t^2 - 2t - 2c^2 + c - 2 \leq 0 \quad p(c, t)$$

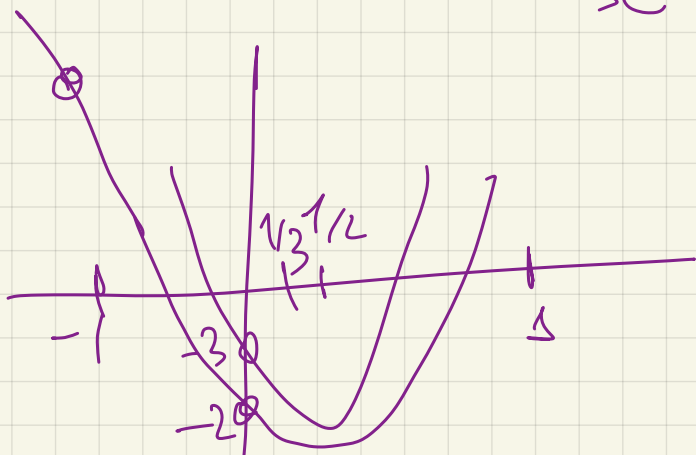
$$c-2t = 3c^2 - 3t^2 + 3$$

$$3t^2 - 2t - 3c^2 + c - 3 = 0 \quad q(c, t)$$

$$-2c^2 + c - 2 \leq -3$$

$$-2 + 1 - 2 \leq -3$$

$$-3c^2 + c - 3 \leq -5$$



$$t_{a1} = \frac{1}{2}$$

$$t_{a2} = \frac{2}{6} = \frac{1}{3}$$

$$\begin{cases} p(-1) \geq 0 \\ q(-1) \geq 0 \end{cases}$$

$$4 + 2 - 2 - 2c^2 + c \geq 0$$

$$3 + 2 - 2 - 3c^2 + c \geq 0$$

$$\begin{cases} 2c^2 - c - 2 \leq 0 \\ 3c^2 - c - 2 \leq 0 \end{cases}$$

$$\frac{1 \pm \sqrt{1+16}}{4}$$

$$\frac{1 \pm \sqrt{17}}{4}$$

$$c \in \left(\frac{1-\sqrt{17}}{4} ; \frac{1+\sqrt{17}}{4} \right)$$

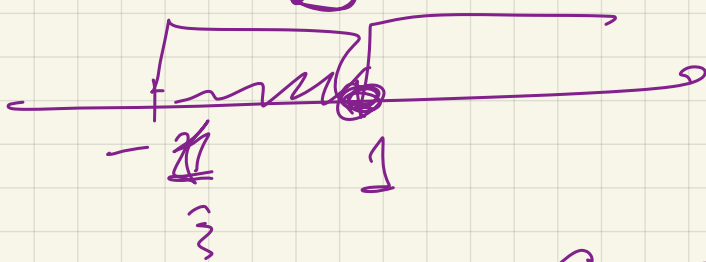
$$c \geq 1$$

$$\frac{1 \pm \sqrt{1+24}}{6}$$

$$= \frac{1 \pm 5}{6}$$

$$c \in \left[1 ; \frac{1+\sqrt{17}}{4} \right)$$

$$c = 1$$



$$c = 1$$

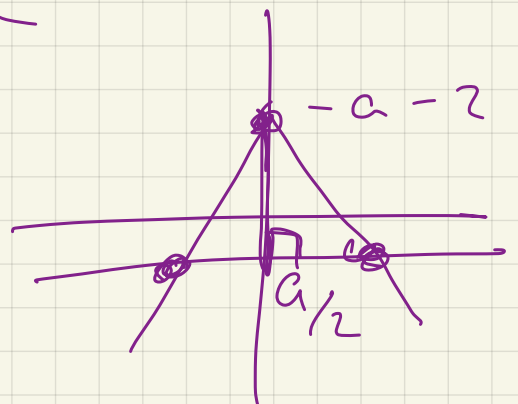
$$a = -1$$

$$a = -1$$

$$\frac{3a}{2} < -2$$

$$\frac{a}{2} < -a-2$$

$$y = a|x-2| - a-2$$



$$A: y = a|x-2| + |a|-2$$

$$y = \frac{a}{2}$$

unknown

$$S \leq \frac{1}{2}$$

$$a = 1$$

$$y = 1|x-2| + 1-2 = |x-2|-1$$

$$a = 0$$

$$y = -2$$

$$y = 0$$

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$$\begin{cases} a > 0 & \textcircled{1} \\ a < 0 & \textcircled{2} \end{cases}$$

$$\textcircled{1} y = a|x-2| + a-2 \Rightarrow$$

$$y = a|x| + a-2$$

$$y = \frac{a}{2}$$

$$a > 0$$

$$A(0; a-2)$$

$$B\left(\frac{a-2}{2a}; \frac{a}{2}\right)$$

$$C\left(\frac{a-4}{2a}; \frac{a}{2}\right)$$

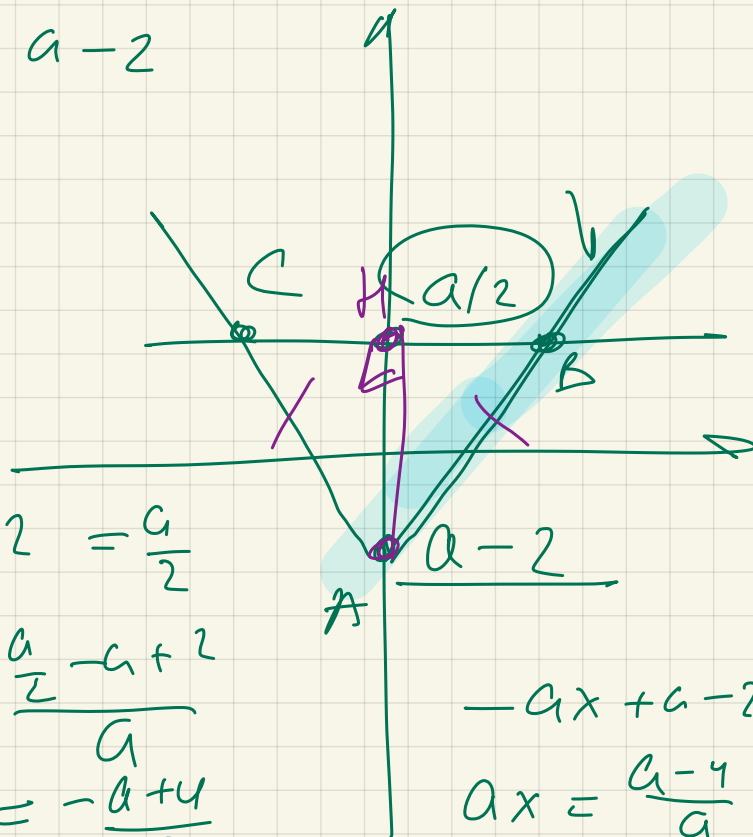
$$ax + a-2 = \frac{a}{2}$$

$$x = \frac{\frac{a}{2} - a + 2}{a}$$

$$x = -\frac{a-4}{2a}$$

$$-ax + a-2 = \frac{a}{2}$$

$$ax = \frac{a-4}{a}$$



$$a - 2 < \frac{a}{2}$$

$$\frac{a}{2} < 2$$

$$0 < a < 4$$

$$S = \frac{AK \cdot CB}{2}$$

$$AK = \frac{a}{2} - (a - 2) = \frac{4 - a}{2}$$

$$CB = -\frac{a - 4}{2a} + \left(\frac{4 - a}{2a}\right) = \frac{-a + 4 + 4 - a}{2a} = \frac{-2a + 8}{2a} = \frac{-a + 4}{a}$$

$$\frac{4 - a}{a}$$

$$S = \frac{1}{2} \left(\frac{4 - a}{2} \right) \left(\frac{4 - a}{a} \right) \leq \frac{1}{2}$$

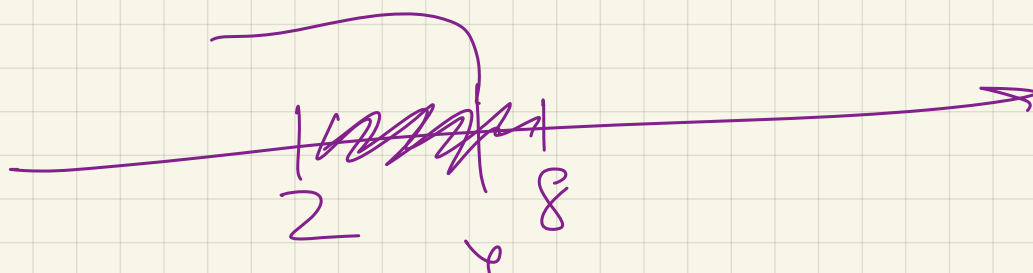
$$\frac{(4 - a)^2}{2a} \leq 1$$

$$\frac{(4 - a)^2}{2a} \leq 2a$$

$$16 - 8a + a^2 - 2a \leq 0$$

$$\frac{a^2 - 10a + 16}{2a} \leq 0$$

$$\frac{10 \pm \sqrt{100 - 64}}{2} \rightarrow \begin{matrix} 8 \\ 2 \end{matrix}$$

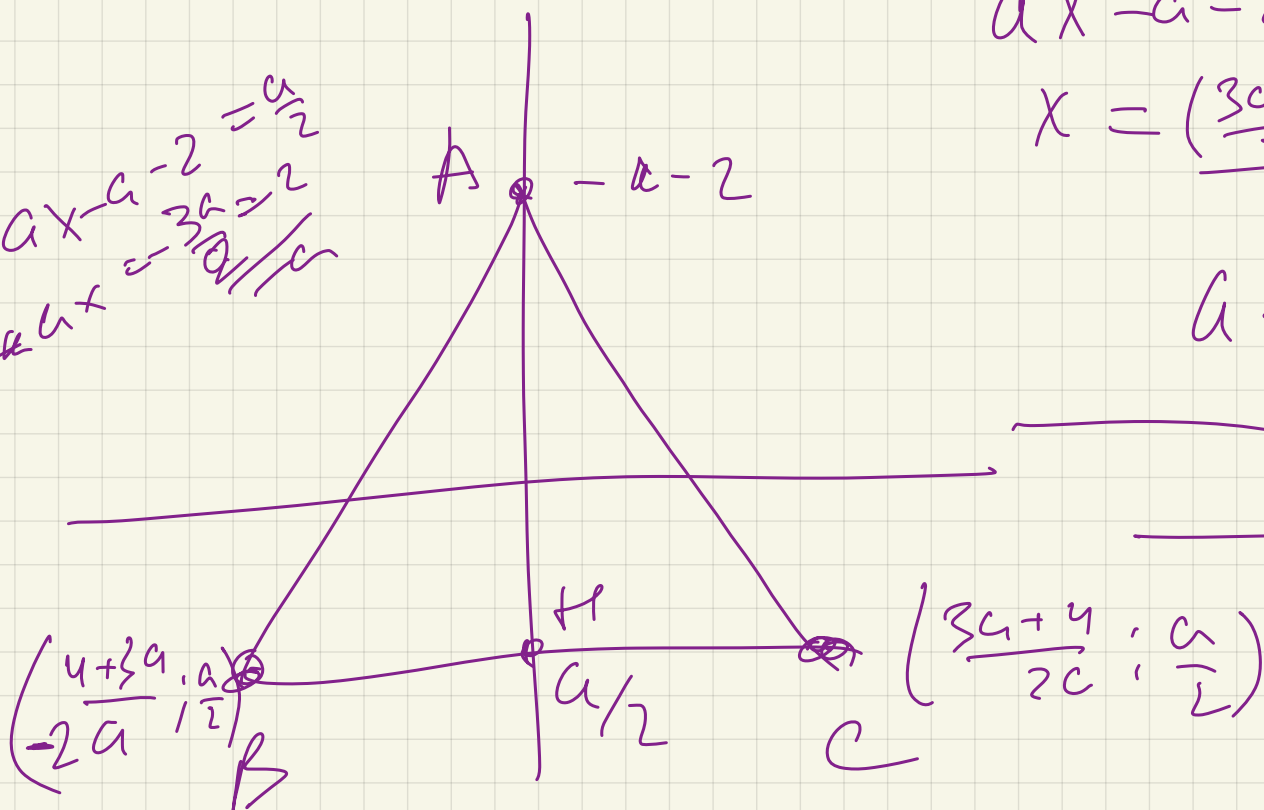


$$a \in [2, 4)$$

$$\begin{aligned} -ax - a - 2 &= \frac{a}{2} \\ 4x &= \frac{3a+4}{a} \end{aligned}$$

$$\begin{aligned} ax - a - 2 &= \frac{a}{2} \\ x &= \frac{\left(\frac{3a}{2} + 2\right)^2}{a} = \frac{3a+4}{2a} \end{aligned}$$

$$a < -\frac{4}{3}$$



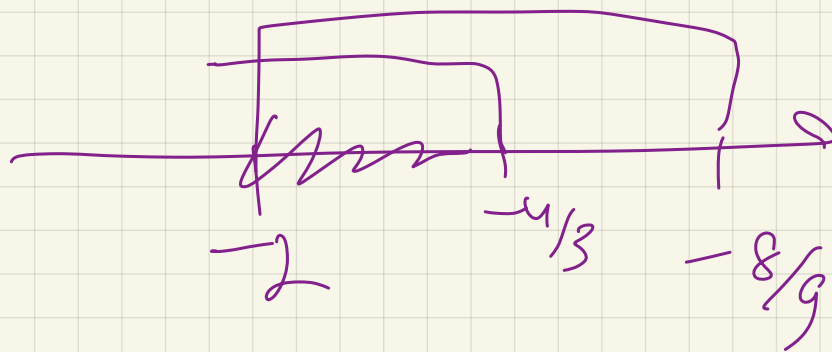
$$\begin{aligned} AH &= -a - 2 - \frac{a}{2} = -\frac{3a}{2} - 2 \\ &= \frac{-3a-4}{2} \end{aligned}$$

$$BC = \frac{-4-3a-4-3a}{2a} = \frac{-6a-8}{2a}$$

$$\begin{cases} S' = \frac{1}{2} \left(\frac{-3a-4}{a} \right) \left(\frac{-3a-4}{2} \right) \leq \frac{1}{2} \\ a < -\frac{4}{3} \end{cases}$$

$$\left(\frac{-3a-4}{2a} \right)^2 \leq 1$$

$$9a^2 + (6 + 4a)a \leq 2a$$



$$[-2; -\frac{4}{3}) \cup [-\frac{8}{9}; 0)$$